Every setting is read off the file itself: no per-dataset tuning

Worked on one real injection (batch4_MT_20210729_003C.mzML): 850,076 centroids in 780 MS1 scans. The same code runs on every file; the sidecar (.params.json) records what the file said.

1 The file

MS1 centroids only
(RT, m/z, intensity)

scan times
→ interval 0.38 s

all intensities
→ floor = 1st percentile
3,896 counts

2 Seeds: the 2,000 strongest ions

strongest first, each ≥ 20 ppm from every earlier seed

each seed: its EIC around the apex (± 20 ppm, 3-point mean)

that is the only input to steps 3 and 4

3 Measured from the seeds

peak width: median FWHM 2.06 s
= 5.4 scans (1,980 seeds measured)

m/z error vs intensity:
$\sigma(I) = \sqrt{(a^2 + b^2/I)}$, a 0.30 ppm, b 82
(44,333 centroids of the seed traces)

scan-to-scan noise vs intensity:
$\sigma(I)^2 = \text{floor}^2 + c \cdot I + r^2 \cdot I^2$, c 2,510, r 0.030
(18,848 residuals, 8 bins)

per-scan gain shared by all ions
(here: none beyond the noise, sd 0.000)

4 Settings the picker runs with

link centroids across K = 3 scans
(K = half the width in scans, 1–3)

m/z tolerance cap 5.4 ppm
(4 σ at the floor intensity, 5–30 ppm)
per pair: the two centroids' own σ(I)

a peak needs ≥ 3 scans
(half the width in scans, 3–5)
smoothing kernel 3 pass(es) of [1 2 1]
(half the median width)

every S/N and valley test in units of the measured noise, not a fixed count

Fixed in the code and the same for every file: the ratios (half the width, 4 σ, 3 σ), the seed count (2,000) and the seed EIC width (20 ppm). Everything with a unit (seconds, ppm, counts) comes from the file.